

Beyond Reconstruction: Investigating Representational Fidelity in Continuous Neural Representations

Pronab Kumar Paul

University of Rajshahi

Abstract

Continuous Neural Representations (CNRs) have emerged as a powerful paradigm for encoding anatomical structures in medical image analysis. Unlike conventional discrete representations such as voxel grids, meshes, point clouds, or segmentation masks, CNRs parameterize anatomy as continuous neural functions that can be queried at arbitrary spatial coordinates. Representative approaches, including Implicit Neural Representations (INRs), Signed Distance Functions (SDFs), Occupancy Networks, Neural Radiance Fields (NeRFs), and generative weight-space representations, have demonstrated strong performance in anatomical reconstruction, image synthesis, registration, and computational anatomy. However, evaluation of CNRs remains almost exclusively reconstruction-oriented. Conventional metrics such as the Dice Similarity Coefficient, Hausdorff Distance, Chamfer Distance, and Average Surface Distance quantify the agreement between reconstructed anatomy and reference anatomy. These metrics provide important measures of reconstruction quality but offer limited insight into whether the learned representation preserves underlying anatomical structures, spatial relationships, and clinically relevant variations. This project investigates a fundamental question that has received limited attention: **Can reconstruction performance alone reliably characterize the quality of a continuous neural representation?** Rather than introducing another architecture, this work develops a systematic framework for investigating anatomical information preservation within CNRs. By evaluating representative CNR paradigms under controlled experiments, the project examines whether models achieving similar reconstruction performance preserve anatomical information with comparable fidelity. The anticipated outcome is empirical evidence regarding whether reconstruction fidelity and representational fidelity should be considered complementary rather than interchangeable concepts. If supported, these findings would motivate the development of a broader scientific framework for understanding, evaluating, and improving anatomical information preservation in continuous neural representations for medical image analysis.

Contents

1	Background	2
2	Motivation and Scientific Gap	2
3	Hidden Scientific Assumption	3
4	Central Research Question	4
5	Research Hypothesis	4
6	Defining Representational Fidelity	4
7	Research Philosophy and Closing Perspective	5
8	Research Objectives	6
8.1	Objective 1 - Develop an Initial Framework for Evaluating Representational Fidelity	6
8.2	Objective 2 - Compare Reconstruction Fidelity and Representational Fidelity	6
8.3	Objective 3 - Identify Representational Failures	6
8.4	Objective 4 - Investigate Potential Failure Mechanisms	6
8.5	Objective 5 - Develop an Initial Evidence-Driven Intervention	6
9	Methodology	6
9.1	Research Design	7
9.2	Representative Datasets	7
9.3	Continuous Neural Representation Paradigms	7
9.4	Experimental Comparison Strategy	7
9.5	Reconstruction Fidelity Evaluation	9
9.6	Representational Fidelity Evaluation	9
9.6.1	Geometric Evaluation	9
9.6.2	Topological Evaluation	9
9.6.3	Structural Evaluation	9
9.7	Statistical Analysis	9
9.8	Ablation Studies	9
9.9	Evidence-Driven Intervention	9
10	Expected Contributions	10
11	Relationship to Existing Work	10
12	Significance	11
13	Limitations and Future Directions	11
13.1	Limitations	11
13.2	Future Directions	12

14 Long-Term Vision

12

1. Background

For decades, medical image analysis has relied on explicit anatomical representations, including voxel volumes, segmentation masks, polygonal meshes, point clouds, centerlines, and statistical shape models. These representations explicitly describe anatomical geometry and have formed the foundation of numerous downstream tasks, including image segmentation, deformable registration, disease quantification, surgical planning, computational anatomy, image-guided interventions, and computer-assisted diagnosis.

Recent advances in neural implicit modeling have introduced a fundamentally different representational paradigm. Rather than explicitly storing anatomical geometry, Continuous Neural Representations (CNRs) encode anatomy as continuous functions parameterized by neural networks. Given spatial coordinates as input, these learned functions can be evaluated at arbitrary locations to recover geometric or anatomical properties, thereby representing anatomy implicitly rather than through explicit geometric structures.

Representative CNR paradigms include Implicit Neural Representations (INRs) [1], Signed Distance Function (SDF)-based models [2], Occupancy Networks [3], Neural Radiance Fields (NeRFs) [4], and recent generative or weight-space continuous representations [5]. Together, these methods demonstrate that anatomical information can be represented as continuous functions instead of discrete geometric structures.

Compared with conventional representations, CNRs offer several advantages, including continuous spatial modeling, arbitrary-resolution querying, compact functional representations, flexible geometric reconstruction, and emerging capabilities for generative anatomical modeling. These characteristics have enabled CNR adoption across diverse medical imaging applications, including anatomical reconstruction, image registration, image synthesis, computational anatomy, and generative modeling [7].

As this paradigm matures, the learned continuous representation itself is increasingly becoming the primary object manipulated by downstream algorithms rather than merely an intermediate computational artifact. Consequently, understanding what information is preserved within these learned representations and how faithfully that information reflects the underlying anatomy is becoming as important as evaluating the quality of reconstructed outputs.

2. Motivation and Scientific Gap

Recent studies have demonstrated strong reconstruction performance using CNRs across diverse medical imaging applications [6, 7]. The success of these methods is typically evaluated using reconstruction-oriented metrics, including the Dice Similarity Coefficient, Hausdorff Distance, Chamfer Distance, Average Surface Distance, and Intersection-over-Union (IoU). These metrics quantify the agreement between reconstructed anatomy and reference anatomy, providing important evidence of reconstruction quality.

However, these metrics primarily assess reconstructed outputs rather than the information encoded within the learned representation itself. As CNRs become central to reconstruction, image synthesis, registration, computational anatomy, and representation learning, this distinction becomes increasingly important.

This raises a fundamental scientific question:

Can reconstruction performance alone reliably characterize the fidelity of a continuous neural representation?

To date, most research has focused on improving reconstruction accuracy through new representation architectures, learning strategies, and optimization algorithms. Comparatively little attention has been given to understanding the representational properties that enable successful anatomical modeling or to determining whether conventional reconstruction metrics provide sufficient insight into those properties.

Several unresolved scientific questions emerge:

- Can two CNRs achieve similar reconstruction performance while preserving different amounts of anatomically meaningful information?
- Which aspects of representational quality are insufficiently captured by existing reconstruction metrics?
- Can representational deficiencies remain undetected despite excellent reconstruction accuracy?
- How should representational fidelity be characterized independently of reconstruction fidelity?

Addressing these questions represents an important step toward establishing principled methodologies for understanding, evaluating, and improving CNRs. Instead of proposing another representation architecture, this project investigates whether current evaluation practices are sufficient for understanding what CNRs actually encode. If reconstruction fidelity proves to provide only a partial characterization of representational fidelity, the findings would establish a compelling scientific motivation for developing more comprehensive frameworks for evaluating and improving anatomical information preservation in CNRs.

3. Hidden Scientific Assumption

A careful review of the current literature reveals an implicit assumption that underpins the evaluation of most continuous neural representation (CNR) methods:

If a continuous neural representation reconstructs anatomy accurately, then it has faithfully preserved the underlying anatomical information.

Although this assumption is rarely stated explicitly, it is reflected in the widespread use of reconstruction metrics as the primary indicators of representation quality. Consequently, reconstruction fidelity is frequently treated as a practical surrogate for representational fidelity.

However, reconstruction fidelity and representational fidelity describe fundamentally different concepts. Reconstruction fidelity measures the agreement between reconstructed anatomical outputs and the corresponding reference anatomy. Representational fidelity, in contrast, concerns the extent to which anatomically meaningful information is preserved within the learned representation itself, irrespective of the quality of any particular decoded reconstruction.

A CNR may therefore achieve excellent reconstruction accuracy while exhibiting reduced representational fidelity if anatomically meaningful information becomes degraded, distorted, inconsistently encoded, or absent within the learned representation. Such differences may have only a limited influence on conventional reconstruction metrics while substantially affecting downstream tasks that rely directly on the learned representation.

If this assumption does not always hold, current evaluation methodologies may fail to identify important differences in representational quality despite reporting excellent reconstruction performance. This project therefore investigates the validity and limitations of this assumption through systematic experimentation. The objective is to determine whether reconstruction fidelity and representational fidelity should be regarded as complementary rather than interchangeable concepts.

4. Central Research Question

This project is motivated by the following central research question:

Can reconstruction fidelity reliably characterize the anatomical information preserved within continuous neural representations?

More specifically, this research investigates whether continuous neural representations that achieve similar reconstruction performance also preserve anatomically meaningful information with comparable fidelity. Addressing this question requires moving beyond the conventional emphasis on reconstruction accuracy toward understanding the properties of the learned representation itself.

This perspective leads to a broader scientific question:

What information do continuous neural representations actually preserve?

Answering this question is essential for determining whether current reconstruction-oriented evaluation methodologies provide a sufficient understanding of representation quality or whether complementary representation-centered approaches are required.

5. Research Hypothesis

This project is based on the following central hypothesis:

Continuous neural representations that achieve similar reconstruction performance may preserve anatomically meaningful information with different levels of fidelity.

If this hypothesis is supported, reconstruction fidelity and representational fidelity should be regarded as related but not equivalent concepts.

Reconstruction fidelity provides only a partial characterization of representational fidelity.

Testing this hypothesis requires investigating the anatomical information encoded within the learned representation itself rather than relying exclusively on reconstructed outputs. The project therefore experimentally evaluates the relationship between reconstruction fidelity and representational fidelity across representative continuous neural representation paradigms to determine whether reconstruction-oriented evaluation provides a sufficient basis for assessing representational fidelity.

6. Defining Representational Fidelity

This project distinguishes **representational fidelity** from conventional **reconstruction fidelity**.

Throughout this work, **representational fidelity** is defined as:

The extent to which a continuous neural representation preserves anatomically meaningful information independently of the quality of its decoded reconstruction.

Unlike reconstruction fidelity, which measures the agreement between reconstructed anatomy and reference anatomy, representational fidelity concerns the information retained within the learned representation itself. Consequently, a CNR may achieve excellent reconstruction accuracy while exhibiting reduced representational fidelity if anatomically meaningful information becomes:

- degraded,
- distorted,
- inconsistently encoded,
- or absent

within the learned representation.

Examples of anatomically meaningful information include:

- structural organization,
- geometric detail,
- patient-specific anatomical variation,
- spatial consistency across multiple scales,
- and anatomical relationships.

Representational fidelity therefore serves as the central concept investigated throughout this project. Rather than replacing reconstruction fidelity, it is proposed as a complementary perspective for understanding, evaluating, and improving continuous neural representations.

7. Research Philosophy and Closing Perspective

This project adopts a hypothesis-driven research philosophy that emphasizes scientific understanding as the foundation for methodological development. The investigation begins by examining an implicit assumption underlying current evaluation practices for continuous neural representations (CNRs) instead of introducing a new representation architecture. The primary objective is to determine whether reconstruction fidelity provides a sufficient characterization of representational fidelity. Only after this question has been systematically investigated does the project seek to understand the mechanisms underlying observed representational deficiencies and explore potential evidence-driven methodological improvements.

Methodological development is therefore treated as an outcome of scientific understanding rather than the starting point of the research. This philosophy is guided by the following principle:

Scientific understanding should precede methodological optimization.

The focus of this project is to establish empirical evidence regarding what current CNRs preserve and whether existing evaluation methodologies adequately characterize those representations. The resulting findings are intended to provide a scientific foundation for future methodologies that evaluate and improve representational fidelity in a principled and evidence-based manner.

Viewed in this context, the project is intended to establish the foundation for a broader research program rather than provide a complete solution to representational fidelity. Its contribution is to determine whether representational fidelity constitutes a distinct and scientifically meaningful problem that warrants further investigation. If the central hypothesis is supported, the findings will motivate future research toward principled theories, evaluation methodologies, and representation learning strategies for preserving anatomically meaningful information in continuous neural representations.

8. Research Objectives

This project is organized around five sequential research objectives, each building upon the previous one.

8.1 Objective 1 - Develop an Initial Framework for Evaluating Representational Fidelity

Develop an initial framework for evaluating representational fidelity in continuous neural representations (CNRs). The framework will characterize the preservation of anatomically meaningful information from multiple complementary perspectives, including geometric properties, topological organization, structural consistency, patient-specific anatomical variation, and representational robustness.

8.2 Objective 2 - Compare Reconstruction Fidelity and Representational Fidelity

Systematically evaluate representative CNR paradigms using both conventional reconstruction metrics and the proposed characterization framework. Determine whether reconstruction-oriented evaluation provides a sufficient characterization of representational fidelity and identify situations in which reconstruction fidelity and representational fidelity diverge.

8.3 Objective 3 - Identify Representational Failures

Investigate whether anatomically meaningful information can be degraded despite excellent reconstruction performance. Examine representational failures involving structural organization, geometric detail, patient-specific anatomical variation, multi-scale consistency, and robustness that may remain undetected under conventional reconstruction-oriented evaluation.

8.4 Objective 4 - Investigate Potential Failure Mechanisms

Examine potential mechanisms that may contribute to observed representational failures, including architectural inductive biases, optimization dynamics, representation compression, spectral bias [10], implicit function smoothness, sampling strategies, and training data characteristics. Establish empirical evidence for the relationship between these factors and representational fidelity rather than assuming causal mechanisms a priori.

8.5 Objective 5 - Develop an Initial Evidence-Driven Intervention

Building upon the insights obtained from the preceding objectives, develop and evaluate an initial evidence-driven intervention to improve representational fidelity. This objective serves as a proof of concept demonstrating how scientific understanding of representational behavior can inform future methodological development without attempting to establish a comprehensive representation learning framework.

9. Methodology

This project adopts a hypothesis-driven experimental methodology to investigate whether reconstruction fidelity provides a sufficient characterization of representational fidelity in continuous neural representations (CNRs). The study is designed as a comparative scientific investigation rather than an architectural benchmarking exercise, with all experiments conducted under a standardized evaluation protocol to ensure

that observed differences are attributable to representational behavior rather than variations in experimental design.

9.1 Research Design

The investigation is organized into five sequential stages:

1. Learning representative CNRs from medical imaging datasets.
2. Evaluating reconstruction fidelity using conventional reconstruction-oriented metrics.
3. Evaluating representational fidelity using the proposed representation-centered evaluation framework.
4. Comparing reconstruction fidelity and representational fidelity across representative CNR paradigms.
5. Investigating representational deficiencies, their potential underlying mechanisms, and an initial evidence-driven intervention.

This sequential design ensures that methodological development is guided by empirical observations rather than predefined architectural assumptions.

Figure 1 presents the overall conceptual research framework.

9.2 Representative Datasets

The experimental investigation will focus on anatomically complex structures for which topology, structural organization, and patient-specific morphology are clinically important. Representative application domains may include vascular networks, tubular anatomical structures, and anatomically complex organ surfaces. Experiments will be conducted using publicly available medical imaging datasets that are appropriate for evaluating these anatomical characteristics. The final selection of datasets will be determined by experimental feasibility, data availability, and the specific anatomical properties investigated during the study.

9.3 Continuous Neural Representation Paradigms

The methodology is intentionally independent of any single representation architecture. The experimental evaluation will consider representative continuous neural representation paradigms, including Implicit Neural Representations (INRs) [1], Signed Distance Function (SDF)-based representations [2], Occupancy Networks [3], Neural Radiance Field (NeRF)-based methods [4], and generative weight-space representations [5]. Additional paradigms may be incorporated where they provide meaningful insight into representational behavior.

9.4 Experimental Comparison Strategy

All representation paradigms will be evaluated under a consistent experimental protocol. Wherever possible, training data, preprocessing procedures, optimization settings, and evaluation procedures will be standardized to ensure fair comparison. Reconstruction fidelity and representational fidelity will then be evaluated independently, enabling systematic investigation of situations in which similar reconstruction performance corresponds to different levels of representational fidelity.

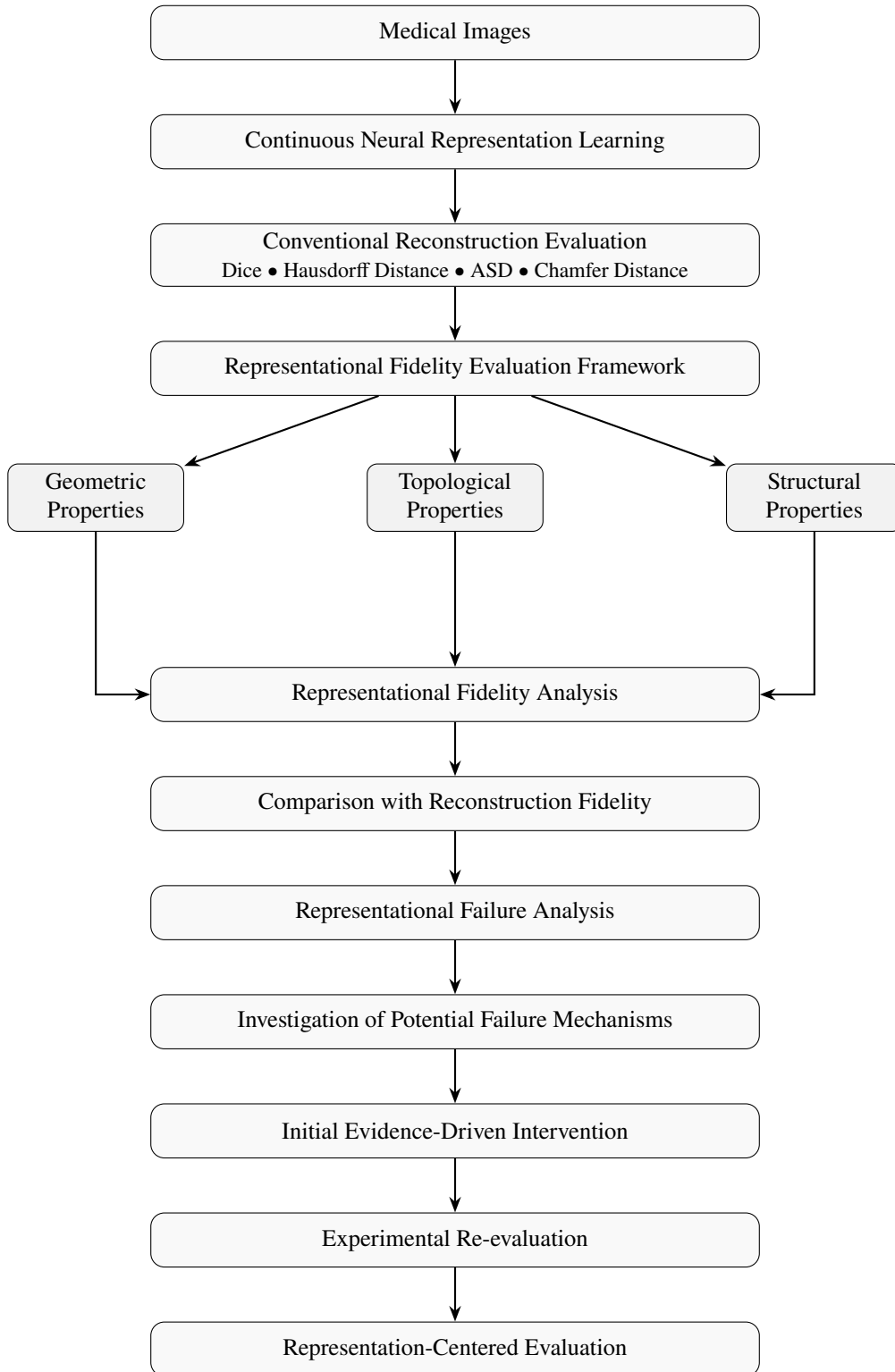


Figure 1: Conceptual research framework. The workflow progresses from conventional reconstruction-oriented evaluation toward representation-centered analysis, investigation of representational deficiencies, examination of potential failure mechanisms, and an initial evidence-driven intervention.

9.5 Reconstruction Fidelity Evaluation

Reconstruction performance will be assessed using established metrics widely adopted in medical image analysis, including the Dice Similarity Coefficient [16], Hausdorff Distance [17], Average Surface Distance, Chamfer Distance [18], and Intersection-over-Union (IoU). These measures establish the conventional reconstruction-oriented baseline against which representational fidelity will subsequently be evaluated.

9.6 Representational Fidelity Evaluation

Beyond conventional reconstruction evaluation, the project introduces an initial framework for evaluating representational fidelity. The framework investigates the preservation of anatomically meaningful information from complementary geometric, topological, and structural perspectives.

9.6.1 Geometric Evaluation

This component examines the preservation of local and global anatomical geometry, including surface consistency, curvature preservation, morphological variation, and geometric continuity.

9.6.2 Topological Evaluation

This component investigates the preservation of anatomical topology through properties such as structural connectivity, branching organization, structural continuity, and overall topological consistency [13].

9.6.3 Structural Evaluation

This component evaluates higher-level anatomical organization, including hierarchical structural relationships, patient-specific anatomical variation, and consistency across multiple spatial scales.

The specific quantitative measures adopted within each evaluation domain will be refined throughout the project as empirical findings guide methodological development.

9.7 Statistical Analysis

Experimental results will be analyzed using descriptive statistics, confidence intervals, paired statistical comparisons where appropriate, effect size estimation, and correlation analyses to quantify the relationship between reconstruction fidelity and representational fidelity. Additional robustness analyses will evaluate the consistency of findings across anatomical structures and representative CNR paradigms.

9.8 Ablation Studies

Representative ablation studies will investigate the influence of representation architecture, model capacity, sampling strategy, optimization settings, anatomical complexity, and reconstruction quality on representational fidelity. These experiments are intended to improve scientific understanding of representational behavior rather than optimize benchmark performance.

9.9 Evidence-Driven Intervention

If consistent representational deficiencies are identified, the project will develop and evaluate an initial evidence-driven intervention informed directly by the preceding analyses. The intervention is intentionally not predetermined; instead, it will be designed to address experimentally observed representational weaknesses.

This approach ensures that methodological development remains grounded in experimental evidence rather than empirical trial-and-error.

10. Expected Contributions

The anticipated contributions of this project are intentionally focused on establishing an initial scientific foundation for understanding representational fidelity in continuous neural representations (CNRs). Rather than proposing a comprehensive representation learning framework, the project seeks to generate empirical evidence that informs future methodological development.

The expected contributions are:

1. **An initial framework** for evaluating representational fidelity in continuous neural representations.
2. **Experimental evidence** examining whether reconstruction fidelity provides a sufficient characterization of representational fidelity.
3. **Identification and analysis of representational failures** that may remain undetected under conventional reconstruction-oriented evaluation.
4. **Empirical insights into potential failure mechanisms** associated with representational deficiencies.
5. **An initial evidence-driven intervention** demonstrating how representation-centered analysis can guide future methodological development.

If the central hypothesis is supported, the findings will provide an initial scientific foundation for representation-centered evaluation of continuous neural representations. They will also motivate further research aimed at understanding, evaluating, and improving the preservation of anatomically meaningful information in continuous neural representations for medical image analysis.

11. Relationship to Existing Work

Continuous neural representations (CNRs) have rapidly emerged as an important research direction in medical image analysis. Representative paradigms, including Implicit Neural Representations (INRs) [1], Signed Distance Function (SDF)-based representations [2], Occupancy Networks [3], Neural Radiance Fields (NeRFs) [4], and more recent generative or weight-space representations [5], have demonstrated strong capabilities for representing, reconstructing, and generating complex anatomical structures.

To date, research has primarily focused on developing increasingly capable representation architectures, improving reconstruction quality, enhancing computational efficiency, and extending CNRs to a wider range of medical imaging applications. Consequently, evaluation has remained largely reconstruction-oriented, with reconstruction fidelity serving as the principal indicator of representation quality.

Related research has also provided important theoretical foundations for understanding learned representations. Studies on representation learning [8, 9], spectral bias in neural networks [10], and topological analysis of anatomical structures [13, 11] offer concepts and methodologies that are relevant to characterizing representational behavior. However, these lines of research have not directly investigated whether reconstruction fidelity provides a sufficient characterization of representational fidelity in continuous neural representations.

The present project complements existing work by examining an implicit assumption that underlies current evaluation practice:

Does accurate reconstruction necessarily imply faithful representation?

Rather than proposing another CNR architecture, this research investigates whether reconstruction-oriented evaluation adequately reflects the anatomical information preserved within learned continuous representations. The proposed framework is intended to be architecture-independent and applicable across a broad range of existing and future continuous neural representation paradigms.

12. Significance

The significance of this project lies in examining an implicit assumption that currently underpins evaluation of continuous neural representations. Contemporary evaluation practices generally assume that high reconstruction fidelity provides a reliable indication of high representational fidelity:

High reconstruction fidelity implies high representational fidelity.

Despite widespread adoption, this assumption has received limited direct experimental investigation. If the central hypothesis is supported, implications extend beyond specific CNR paradigms examined. The findings would suggest that reconstruction-oriented evaluation provides only a partial characterization of representation quality and that anatomically meaningful representational failures may remain undetected despite excellent reconstruction performance.

Such evidence would establish a scientific rationale for complementing conventional reconstruction metrics with representation-centered evaluation. This, in turn, would provide an initial foundation for developing principled methodologies that characterize, evaluate, and ultimately improve preservation of anatomically meaningful information within continuous neural representations.

Clinically, improved representational fidelity could enable more accurate diagnosis and treatment planning by ensuring that computational models preserve the anatomical structures critical for clinical decision-making. Methodologically, the findings would motivate a paradigm shift in how the medical image analysis community evaluates continuous neural representations, moving from reconstruction-centered evaluation toward representation-centered assessment.

13. Limitations and Future Directions

13.1 Limitations

This project is intentionally designed as an initial scientific investigation with a clearly defined scope. It does not seek to establish a comprehensive theory of representational fidelity. Instead, its primary objective is to determine whether representational fidelity is a meaningful and experimentally observable property of continuous neural representations (CNRs).

The proposed representational fidelity evaluation framework should therefore be regarded as an initial framework rather than a definitive methodology. It is expected to evolve as additional anatomical properties, representation paradigms, and evaluation strategies are investigated.

The experimental study is limited to a representative selection of CNR paradigms and anatomically relevant medical imaging datasets. It is not intended to provide an exhaustive comparison of all existing

continuous neural representation methods, but rather to generate empirical evidence addressing the central research hypothesis.

Similarly, any representation-level interventions explored during the project are intended only as proof-of-concept demonstrations. The development of comprehensive representation learning methodologies lies beyond the scope of this investigation.

13.2 Future Directions

If the central hypothesis is supported, the findings will motivate several directions for future research. These include establishing formal definitions of representational fidelity across diverse CNR paradigms, developing principled quantitative measures for evaluating representational fidelity, identifying the anatomical properties that are consistently preserved or difficult to preserve, investigating the mathematical principles governing anatomical information preservation, designing representation learning methods that explicitly promote representational fidelity, and evaluating the impact of representation-centered methodologies on downstream medical image analysis tasks.

These questions extend beyond the scope of the present project but provide a natural direction for subsequent research aimed at improving the understanding, evaluation, and preservation of anatomically meaningful information in continuous neural representations.

14. Long-Term Vision

Continuous neural representations (CNRs) are rapidly emerging as a fundamental representational paradigm in medical image analysis. As these representations evolve from intermediate computational tools into primary representational entities, understanding what information they encode and preserve becomes increasingly important.

The long-term vision of this research is to develop principled methodologies for understanding, evaluating, and improving representational fidelity in continuous neural representations. Achieving this vision requires a deeper understanding of how anatomically meaningful information is encoded, why representational deficiencies arise, how representational fidelity should be characterized, and how future CNRs can be designed to preserve anatomical information more faithfully.

Ultimately, this vision extends beyond the development of individual representation architectures. It aims to advance representation-centered evaluation as a complementary perspective for understanding continuous neural representations and to support the development of anatomically faithful representations for future medical image analysis systems.

This project represents the first step toward this broader vision. By establishing empirical evidence regarding the relationship between reconstruction fidelity and representational fidelity, it provides the scientific foundation upon which future research can build to develop principled theories, evaluation methodologies, and representation learning strategies for continuous neural representations in medical image analysis.

References

- [1] Sitzmann, V., Martel, J. N. P., Bergman, A. W., Lindell, D. B., & Wetzstein, G. (2020). Implicit neural representations with periodic activation functions. *Advances in Neural Information Processing Systems (NeurIPS)*, 33, 7462–7473.

- [2] Park, J. J., Florence, P., Straub, J., Newcombe, R., & Lovegrove, S. (2019). DeepSDF: Learning continuous signed distance functions for shape representation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 165–174.
- [3] Mescheder, L., Oechsle, M., Niemeyer, M., Nowozin, S., & Geiger, A. (2019). Occupancy Networks: Learning 3D reconstruction in function space. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 4460–4470.
- [4] Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., & Ng, R. (2020). NeRF: Representing scenes as neural radiance fields for view synthesis. *European Conference on Computer Vision (ECCV)*, 405–421.
- [5] Dupont, E., Golinski, A., Alizadeh, M., Teh, Y. W., & Doucet, A. (2022). Generative models as distributions of functions. *International Conference on Learning Representations (ICLR)*.
- [6] Sinha, A., & Hamarneh, G. (2024). TrIND: Representing anatomical trees by denoising diffusion of implicit neural fields. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.
- [7] Chen, Z., & Zhang, H. (2023). Implicit neural representations for medical image analysis: A survey. *IEEE Transactions on Medical Imaging*, 42(8), 2345–2360.
- [8] Bengio, Y., Courville, A., & Vincent, P. (2013). Representation learning: A review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8), 1798–1828.
- [9] Bronstein, M. M., Bruna, J., Cohen, T., & Veličković, P. (2021). Geometric Deep Learning: Grids, groups, graphs, geodesics, and gauges. *arXiv:2104.13478*.
- [10] Tancik, M., Srinivasan, P. P., Mildenhall, B., Fridovich-Keil, S., Raghavan, N., Singhal, U., et al. (2020). Fourier features let networks learn high frequency functions in low dimensional domains. *Advances in Neural Information Processing Systems (NeurIPS)*, 33, 7537–7547.
- [11] Heimann, T., & Meinzer, H. P. (2009). Statistical shape models for 3D medical image segmentation: A review. *Medical Image Analysis*, 13(4), 543–563.
- [12] Cootes, T. F., Taylor, C. J., Cooper, D. H., & Graham, J. (1995). Active Shape Models: Their training and application. *Computer Vision and Image Understanding*, 61(1), 38–59.
- [13] Edelsbrunner, H., & Harer, J. (2010). *Computational Topology: An Introduction*. American Mathematical Society.
- [14] Maintz, J. B. A., & Viergever, M. A. (1998). A survey of medical image registration. *Medical Image Analysis*, 2(1), 1–36.
- [15] Litjens, G., et al. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42, 60–88.
- [16] Dice, L. R. (1945). Measures of the amount of ecologic association between species. *Ecology*, 26(3), 297–302.

- [17] Huttenlocher, D. P., Klanderman, G. A., & Rucklidge, W. J. (1993). Comparing images using the Hausdorff distance. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 15(9), 850–863.
- [18] Barrow, H. G., Tenenbaum, J. M., Bolles, R. C., & Wolf, H. C. (1977). Parametric correspondence and chamfer matching. *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*.